

continuation

Continuation Document

Revision: 33 **Last modified:** 2026-06-01T11:06:09Z **Description:** Sacred invariant resumption document for Helix Cluster OS **Authority:** Constitution §12.10
Maintainer: Operator + AI loop

§0: How to use this document

Any CLI agent resuming work on this project MUST read this file first.

§1: Recently Completed Work (last 10 commits)

Commit	Message
a084eb1	Foundation wave 2: anti-bluff harden 29 pkg test items + §1.1 mutation runner + -race CI gate
3fb815d	Foundation wave 1: mark 30 verified items Completed (HXC-1001..1009/1016-1018/1021/1024/1026-1027/1030-1037/1042-1045/1062-1063)
dbed10c	Foundation wave 1: anti-bluff remediation of etcd/lock/infra/build/security/crypto/jwt/hxcregistry/storage/wasm (verified)
8d7030c	Foundation: add pkg/testing/evidence — §7.1 positive-evidence test helper (TDD + §1.1 mutation-paired)
47b6d91	Phase 0: ingest 614-item phase ledger into hxc_registry.db (HXC-1001..1614); DB-as-source-of-truth
a15c272	Phase 0: wire docs_chain (§11.4.106) — tracked_docs context, engine wrapper, commit gate sync+verify, canonical doc dedup manifest
4554e9a	Phase 0 enabling: integrate constitution upstream (§11.4.106 Docs Chain + workable-items DB), add docs_chain + helixqa submodules, purge stale test binaries, gitignore docs_chain runtime
279bce8	Bump security submodule -> fb1f3ce (Phase 8C e2ee interop + gpuattest spot-check/seal)
f73806e	Phase 8C gaps: model-integrity gate, attestation-gated admission, GPU class model, fiber stake admission, inference proxy (PCS-6/CLAUDE-1)
31c4490	Phase 8B gaps: GPU pool manager, cost broker, burst hysteresis, local GPU TCO, provider adapter (PCS-6/CLAUDE-1)

§2: Environment Snapshot

Property	Value
Branch	main
Commit	a084eb1
Timestamp	2026-06-01T11:06:09Z

§3: Active Work

HXC	Title	Status
HXC-922	Node service, advisory locks, chaos tests, docs	✓ Done
HXC-921	Web UI, K8s/Helm, integration, E2E, benchmarks	✓ Done
HXC-920	Health, security, build, wireguard services	✓ Done
HXC-919	htmux, messaging, GPU, WASM, storage, build manifest	✓ Done
HXC-918	Wire scheduler/session gRPC to real backends, etcd discovery	✓ Done
HXC-916	Stub expansion (config, retry, ratelimit, validator, build cache)	✓ Done
HXC-914	LLM, policy, setup, distributed lock	✓ Done
HXC-913	Health, security, ClassAds parser	✓ Done
HXC-912	Gateway, helixd, helix-agent	✓ Done
HXC-911	Scheduler, Session gRPC services	✓ Done
HXC-910	Console, security, testing infra	✓ Done

§4: Next Planned Work

All MVP, Phase 2, Phase 3, and Phase 4 are **COMPLETE**. Potential future enhancements:

1. **Performance optimization** — Profile and optimize hot paths
2. **Multi-region support** — Cross-region cluster federation
3. **GPU scheduling v2** — Multi-GPU, fractional GPU allocation

4. **Web UI real-time** — WebSocket integration for live updates
5. **Operator pattern** — Kubernetes operator for cluster management

§5: Known Issues / Blockers

- None

§6: Quick Commands

Run all tests (requires GOWORK due to workspace modules)

```
GOWORK=$(pwd)/go.work go test ./... -race -count=1
```

Build all binaries

```
GOWORK=$(pwd)/go.work go build ./cmd/...
```

Build web UI

```
cd web && npm run build
```

Render Helm chart

```
helm template helix-cluster deploy/helm/
```

Run benchmarks

```
go test -bench=. ./test/benchmark
```

Run chaos tests

```
go test -race -count=1 ./test/chaos
```